Operating System Concept

Introduction

Presentation Outline

- Introduction to Operating Systems
- Types of Operating Systems
  - Batch Processing Systems
  - Multiprogramming Systems
  - Time-Sharing Systems
  - GUI-Based Systems
  - Networked Systems
  - Mobile Operating Systems
  - AI-Powered Operating Systems

Presentation Outline

- Operating System Structures / Architectures
  - Simple Architecture
  - Monolithic Architecture
  - Microkernel Architecture
  - Layered Architecture
  - Modular Architecture
- Single User Operating Systems
- Multi-User Operating Systems
  - Distributed Systems
  - Time-Sliced Systems
  - Multiprocessor Systems
- Spooling and Buffering

Learning Objectives

- Define the concept and role of an Operating System.
- Identify and explain major types of Operating Systems.
- Compare Batch, Multiprogramming, Time-Sharing, GUI, Networked, Mobile, and AI-Powered systems.
- Describe different Operating System architectures and their characteristics.
- Differentiate between Single User and Multi-User Operating Systems.
- Explain Distributed, Time-Sliced, and Multiprocessor systems.
- Understand the concepts of Spooling and Buffering in OS environments.
- Analyze the advantages and limitations of various Operating System designs.

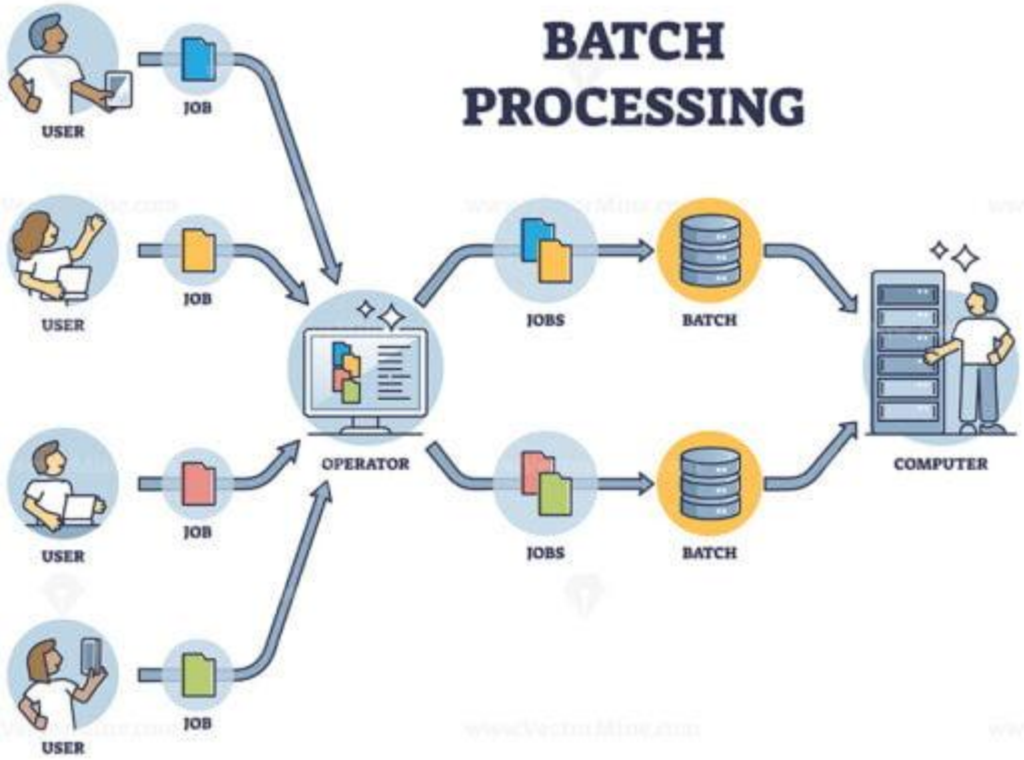
Operating System Types

01 - Batch Processing Systems

- These systems were popular From **1940s to 1950s.**
- The users of a batch operating system **did not interact with the computer directly**.
- Each user prepared **his job on an off-line device like punch cards and submited it to the computer operator** who then batched the similar jobs together to speed up processing and run as a group.
- The programmers left their programs with the operator and the operator then sorted the programs with similar requirements into batches.
- In such systems, CPU usage was very low and it was difficult to prioritize jobs over one another.

Operating System Types

01 - Batch Processing Systems



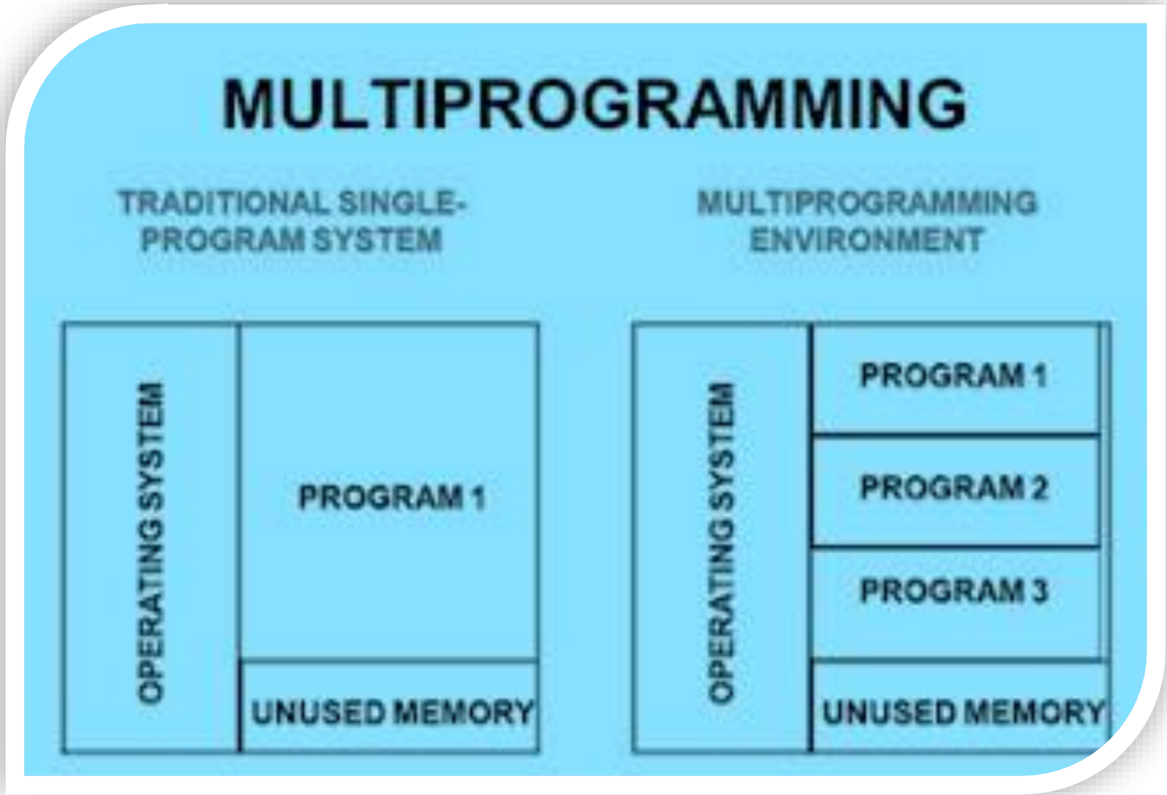
Operating System Types

02 - Multiprogramming Systems

- Emerged during the **1950s and 1960s** and transformed computer systems.
- Allowed **multiple programs to be loaded into main memory simultaneously**.
- Each program was allocated its **own memory space**.
- Improved **CPU utilization** by reducing idle time.
- When one program waited for an **I/O operation**, the CPU switched to another program.
- Increased overall **system efficiency and throughput.**
- Formed the foundation for modern **multitasking operating systems**.

Operating System Types

02 - Multiprogramming Systems



Operating System Types

03- Time Sharing Systems

- Developed and widely used during the **1960s and 1970s**.
- Time-sharing is a **logical extension of multiprogramming**.
- Allows **multiple users to access a computer system simultaneously.**
- CPU time is divided into small units called **time slices (time quantum)**.
- Each user or process receives a **small portion of CPU time.**
- The operating system rapidly switches the CPU among users and processes.
- Uses **CPU scheduling algorithms** to allocate processor time efficiently.

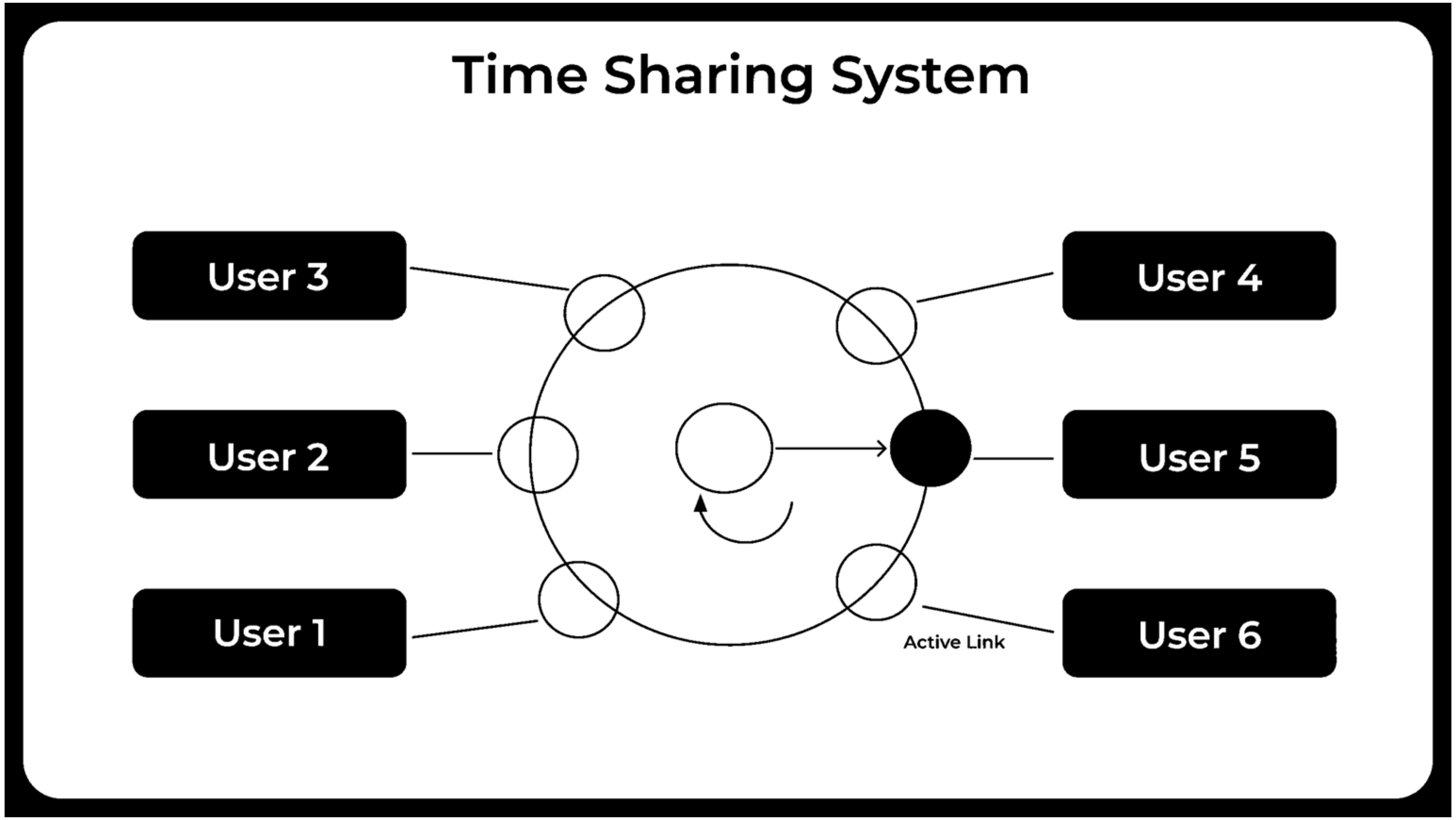
Operating System Types

03- Time Sharing Systems – Conti…

- Employs **multiprogramming techniques** to keep several programs in memory at the same time.
- Provides an **interactive computing environment** for users.
- Gives users the impression that they have **exclusive access to the computer.**
- Improves **CPU utilization** by minimizing processor idle time.
- Enables efficient **resource sharing** among multiple users.
- Offers **quick response times** for user commands and applications.
- Many computer systems that were originally designed as **batch processing systems** were later modified into **time-sharing systems**.
- Common examples include **UNIX and Linux.**

Operating System Types

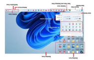
03- Time Sharing Systems – Conti...



Operating System Types

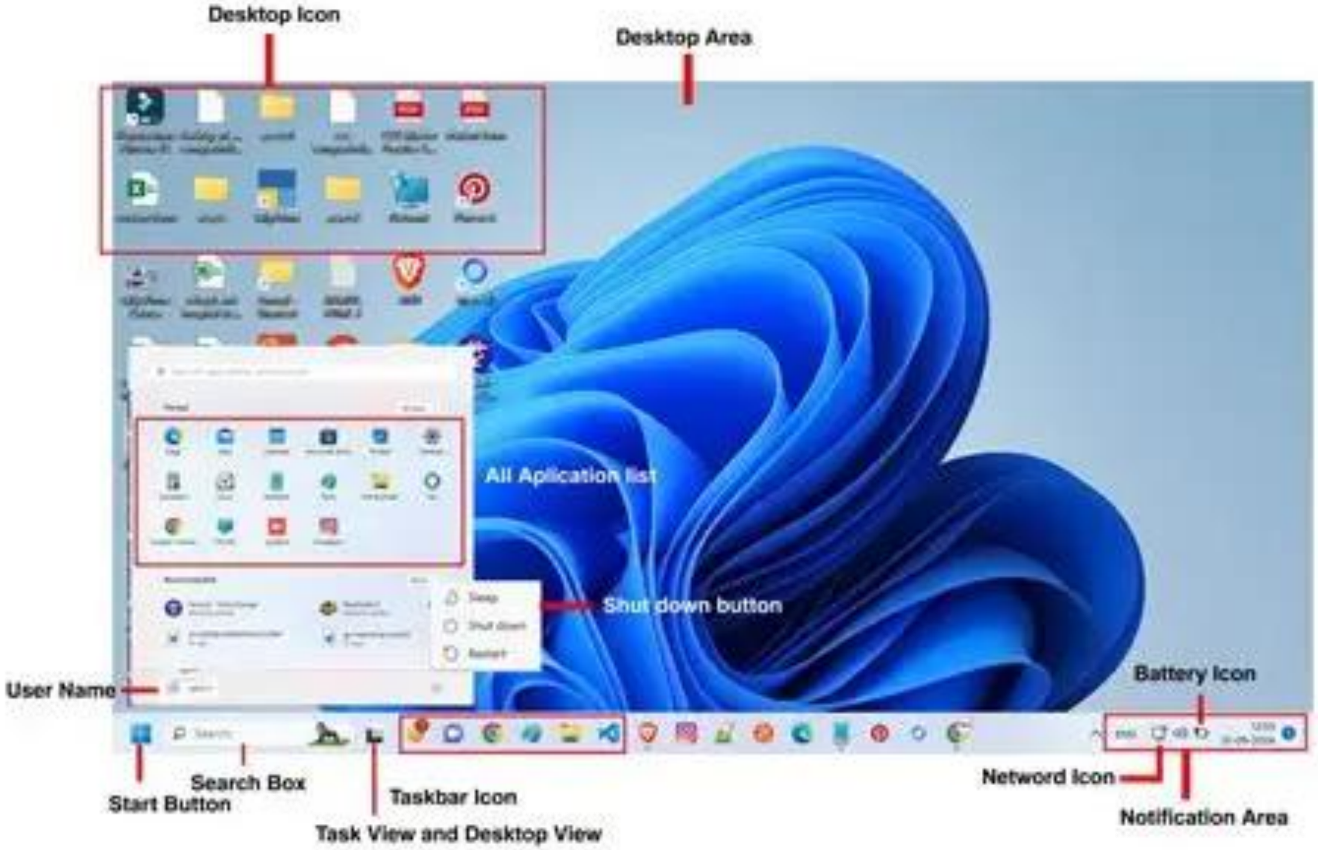
04- GUI Based Systems

- Became popular during the **1970s and 1980s.**
- Provide a **Graphical User Interface (GUI)** for user interaction.
- Users interact through **windows, icons, menus, and buttons**.
- Eliminate the need to memorize and type complex commands.
- Make computers **easier and more user-friendly** for beginners.
- Use pointing devices such as a **mouse or touchpad** for navigation.
- Support **multitasking** by allowing multiple windows and applications to run simultaneously.
- **Microsoft Windows** is one of the most popular GUI-based operating systems and continues to dominate the personal computer market.



Operating System Types

04- GUI Based Systems



Operating System Types

05- Networked Systems

- Became widely popular during the **1980s and 1990s** with the growth of computer networking.
- A **Network Operating System (NOS)** runs on a server and manages network resources.
- Supports the management of **users, groups, files, applications, and security.**
- Allows multiple computers to **share resources** over a network.
- Provides **shared file access** among connected computers.
- Enables **shared printer access**, reducing hardware costs.
- Supports communication and resource sharing within a **Local Area Network (LAN)** or other networks.
- Improves **centralized administration, security, and network management**.

Operating System Types

05- Networked Systems

Operating System Types

06- Mobile Operating Systems

- During the **late 1990s and early 2000s**, **Symbian OS** and **Java ME** were among the most popular operating systems for mobile devices.
- These operating systems were primarily designed for basic mobile functions such as **calling, messaging, and simple applications.**
- The rapid growth of **smartphone technology** increased the demand for more advanced and feature-rich operating systems.
- Modern smartphones require support for **multitasking, internet access, multimedia, touch-screen interfaces, and mobile applications.**
- This demand led to the development of **Android** and **iOS**, the two dominant mobile operating systems today.
- **Android** provides an open-source platform used by many smartphone manufacturers worldwide.
- **iOS**, developed by Apple, offers a secure and highly optimized environment for Apple devices.
- **Both Android and iOS continue to evolve with new technologies, enhanced security, improved performance, and advanced features.**

Operating System Types

06- Mobile Operating Systems

Mobile Operating System

Operating System Types

07- AI Powered

- Since the **2010s**, Artificial Intelligence (AI) has become an integral part of modern operating systems.
- AI enables systems to **understand, learn, and respond** to user behavior and commands.
- Voice assistants such as **Siri, Google Assistant, and Amazon Alexa** use AI technologies to interact with users.
- Modern operating systems can **recognize speech and process natural language commands**.
- AI helps automate tasks such as **scheduling, searching, navigation, and device control**.
- Intelligent features improve **user experience, productivity, and accessibility**.
- AI-powered systems can **analyze user preferences** and provide personalized recommendations.
- The integration of AI continues to make operating systems **smarter, more efficient, and user-friendly.**

Operating System Structures / Architecture.

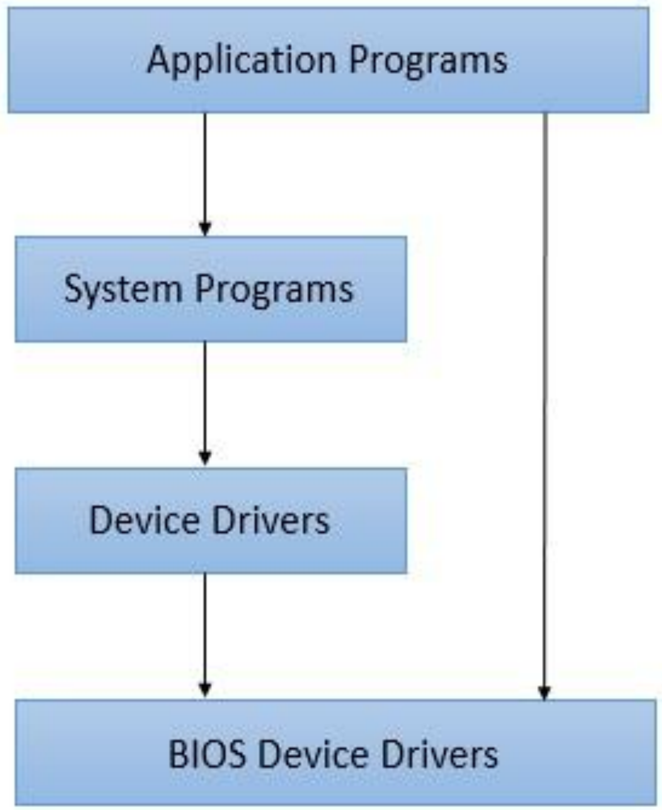
- An operating system allows the user application programs to interact with the system hardware.
- Since the operating system is such a complex structure, its **architecture** plays an important role in its usage.
- Each component of the Operating System Architecture should be well defined with clear inputs, outputs and functions.

Popular Structure / Architectures

1. **Simple Architecture**
2. **Monolith Architecture**
3. **Micro-Kernel Architecture**
4. **Layered Architecture**
5. **Modular Architecture**

Simple Architecture

- There are many operating systems that have a rather simple structure.
- These started as small systems and rapidly expanded much further than their scope.
- A common example of this is MS-DOS.
- It was designed simply for a small amount for people.
- There was no indication that it would become so popular.



Advantages

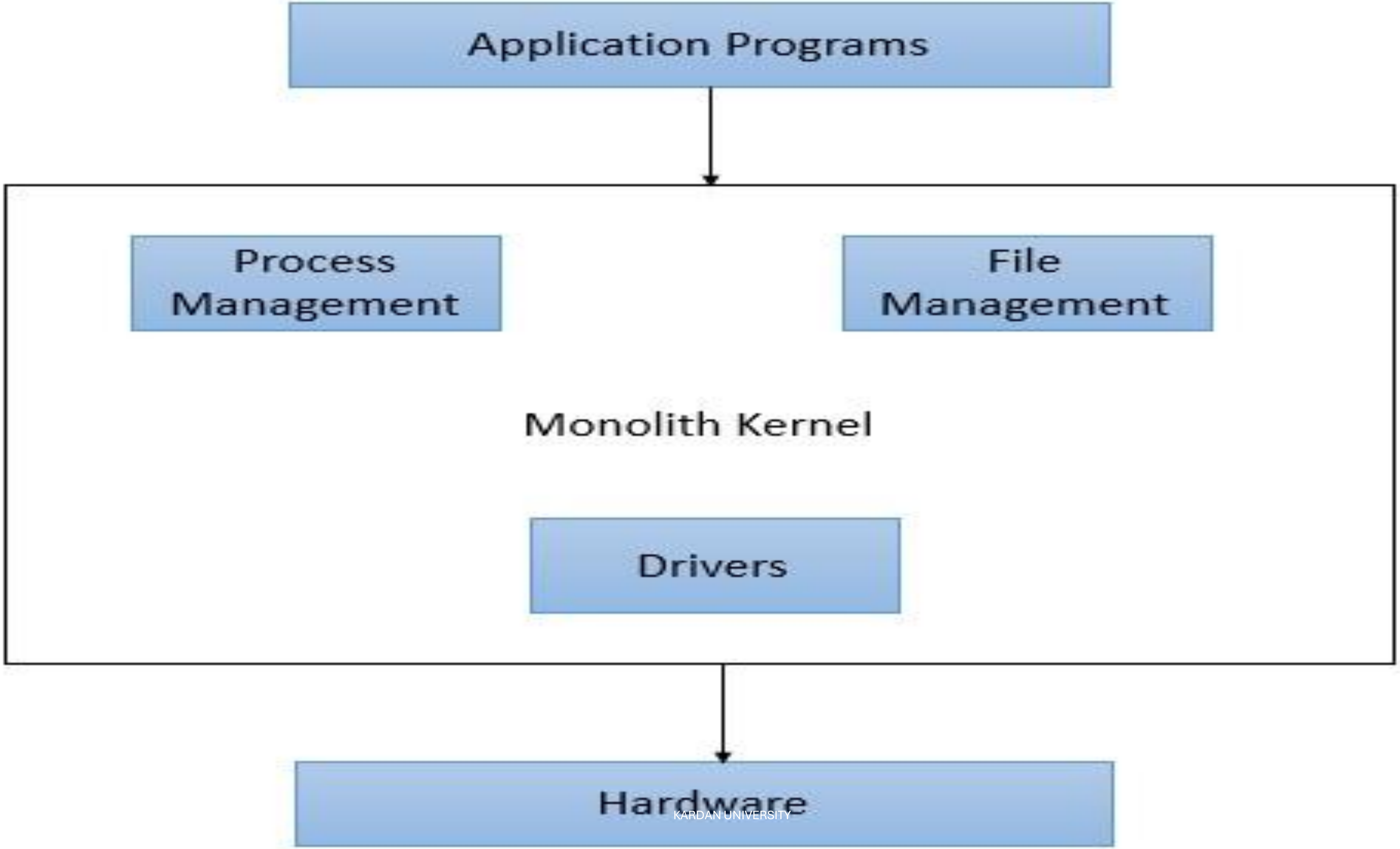
- **Easy Development** - In simple operation system, being very few interfaces, development is easy especially when only limited functionalities are to be delivered.
- **Better Performance** - Such a system, as have few layers and directly interacts with hardware, can provide a better performance as compared to other types of operating systems.

Disadvantages

- **Frequent System Failures** - Being poorly designed, such a system is not strong. If one program fails, entire operating system crashes. Thus system failures are quiet frequent in simple operating systems.
- **Poor Maintainability** - As all layers of operating systems are tightly coupled, change in one layer can impact other layers heavily and making code unmanageable over a period of time.

Monolithic Architecture

- In monolithic architecture operating system, a **central piece of code called kernel** is responsible for all major operations of an operating system.
- Such operations includes **file management, memory management, device management** and so on.
- The **kernel** is the main component of an operating system and it provides all the services of an operating system to the application programs and system programs.
- The kernel has access to the **all the resources** and it acts as an **interface** with application programs and the underlying hardware.



Application Programs
Process Management
File Management
Monolith Kernel
Drivers
Hardware

Advantages

- **Easy Development** - As kernel is the only layer to develop with all major functionalities, it is easier to design and develop.
- **Performance** - As Kernel is responsible for memory management, other operations and have direct access to the hardware, it performs better.

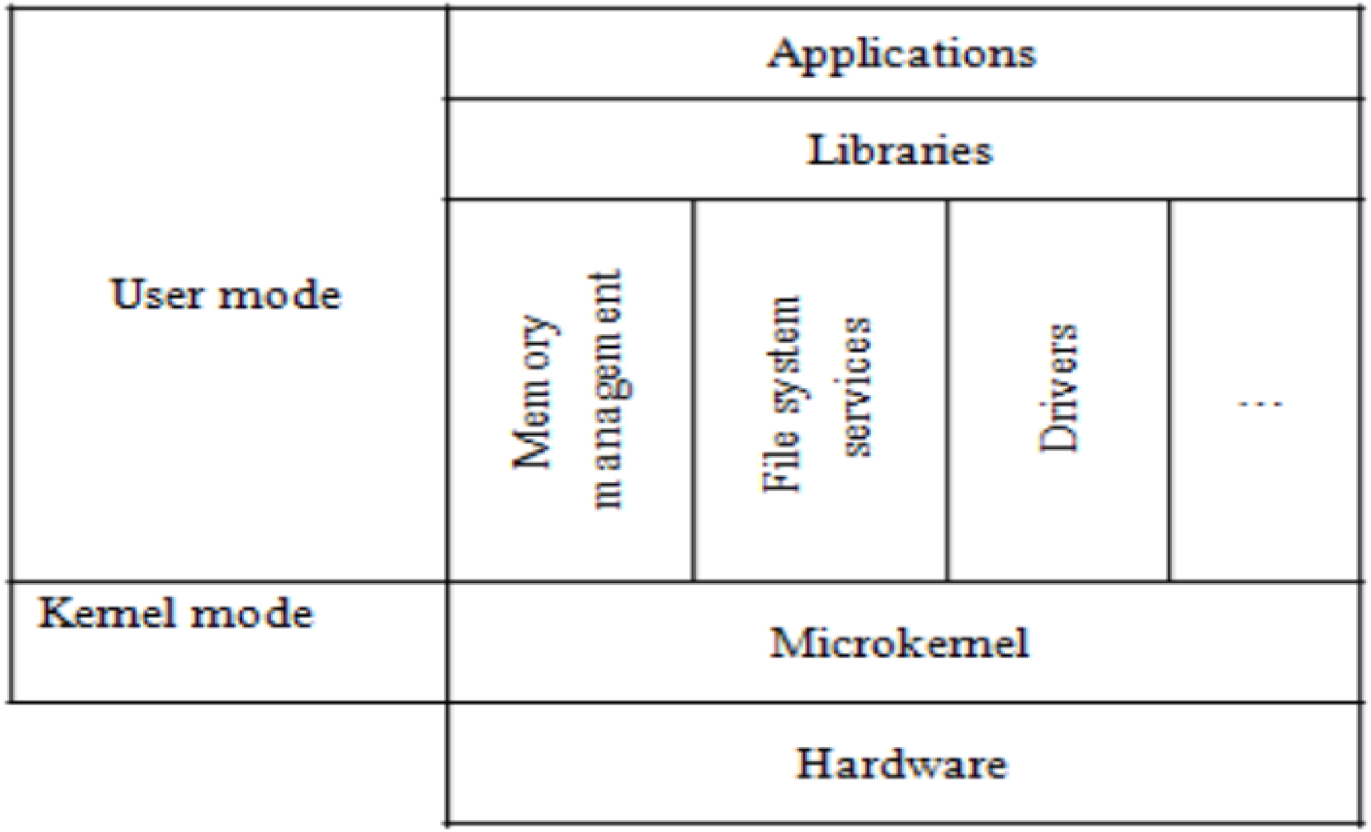
Disadvantages

- **Crash** - As Kernel is responsible for all functions, if one function fails entire operating system fails.
- **Difficult to enhance** - It is very difficult to add a new service without impacting other services of a monolith operating system.

Micro-Kernel Architecture

- As in case monolith architecture, there was single kernel, in micro-kernel, we have multiple kernels each one specialized in particular service.
- Each microkernel is developed independent to the other one and makes system more stable.
- If one kernel fails the operating system will keep working with other kernel's functionalities.

Structure of Micro Kernel



Structure of Microkernel

Advantages

- Reliable and Stable - As multiple kernels are working simultaneously, chances of failure of operating system is very less. If one functionality is down, operating system can still provide other functionalities using stable kernels.

- Maintainability - Being small sized kernels, code size is maintainable. One can enhance a microkernel code base without impacting other microkernel code base.

Disadvantages

- **Complex to Design** - Such a microkernel based architecture is difficult to design.
- **Performance Degradation** - Multi kernel, Multi-modular communication may hamper the performance as compared to monolith architecture.